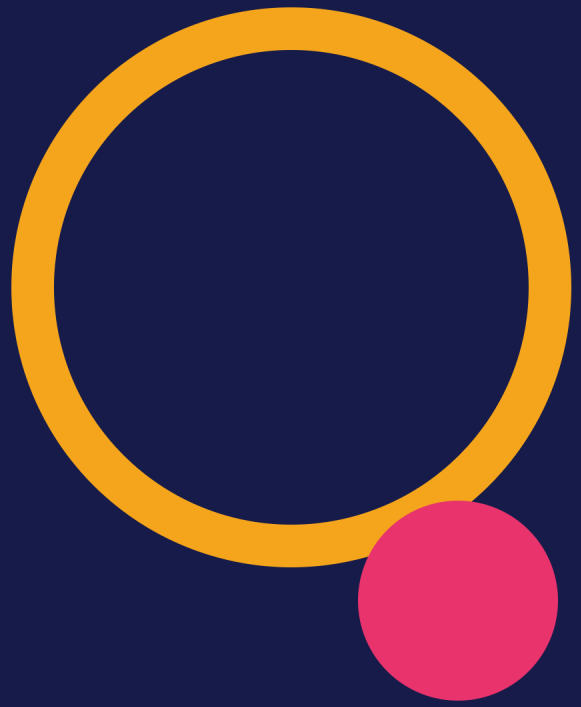
SIH26242 · MINISTRY OF SKILL DEVELOPMENT AND ENTREPRENEURSHIP

Hunar हुनर

AI-assisted Recognition of Prior Learning for India's informally trained skilled workers

AI suggests. The assessor decides.



Hunar – Tanishq Khandelwal – UPES

THE PROBLEM

Skilled, but not recognised

73.1%

of workers outside agriculture work in informal-sector enterprises

58.2%

of regular wage employees outside agriculture have no written job contract

Most learned their trade on the job, so their skill has no certificate. The RPL scheme can recognise it, but practical assessment is manual, slow and inconsistent between assessors and locations.

Source: PLFS Annual Report, Jan–Dec 2025, NSO, MoSPI, Statements 9 and 10

THE CHALLENGE

What SIH26242 asks for, and our answer

The problem statement asks for	Hunar
A worker's self-declaration mapped to the right qualification pack	Multilingual matching against official NSQF performance criteria, showing matched sentences and skill gaps
Guided practical evaluation, with image or video aids	Task checklists, consented photos with faces blurred, and a check for visible safety gear
Standard scoring across assessors and locations	Anchored 0–3 rubrics linked to official skill-standard codes, with a reason for every suggested score
A competency profile and recommendation for assessor sign-off	Results and gaps per skill standard; only the assessor's sign-off finalises the recommendation
Works in low-connectivity settings	Offline-first app that queues every action and syncs in order when the internet returns

OUR SOLUTION

How Hunar works

1

Declare

The worker speaks or types their experience in Hindi, English or Hinglish

2

Match

The closest official NSQF qualification, with matched sentences and gaps

3

Practise

Task checklists; photos taken with consent, faces blurred

4

Check

Is safety gear visible? Was this photo used for another worker?

5

Score

A suggested 0–3 level per criterion, with the reason in plain words

6

Decide

The assessor accepts or overrides (with a reason) and signs off

Works offline, syncs later

Every decision goes into a tamper-evident audit log

STEP 2 · MATCH

From words to an official qualification

Matches every requirement

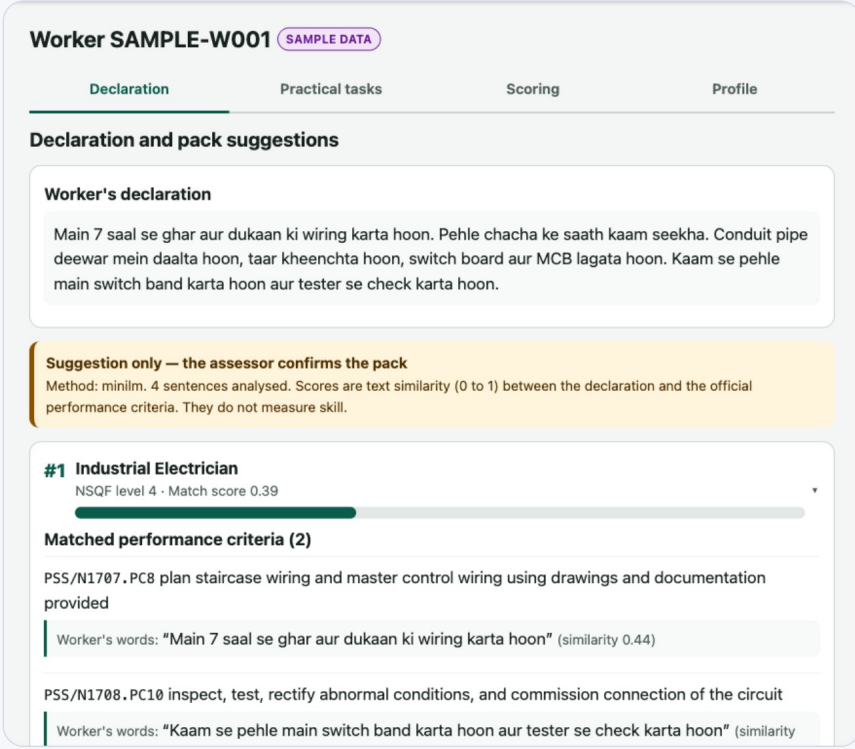
Compares each sentence with every official performance criterion, in Hindi, English or Hinglish

Explains itself

Shows which words matched which criterion, and which criteria had no match: the skill gaps

Built on official data

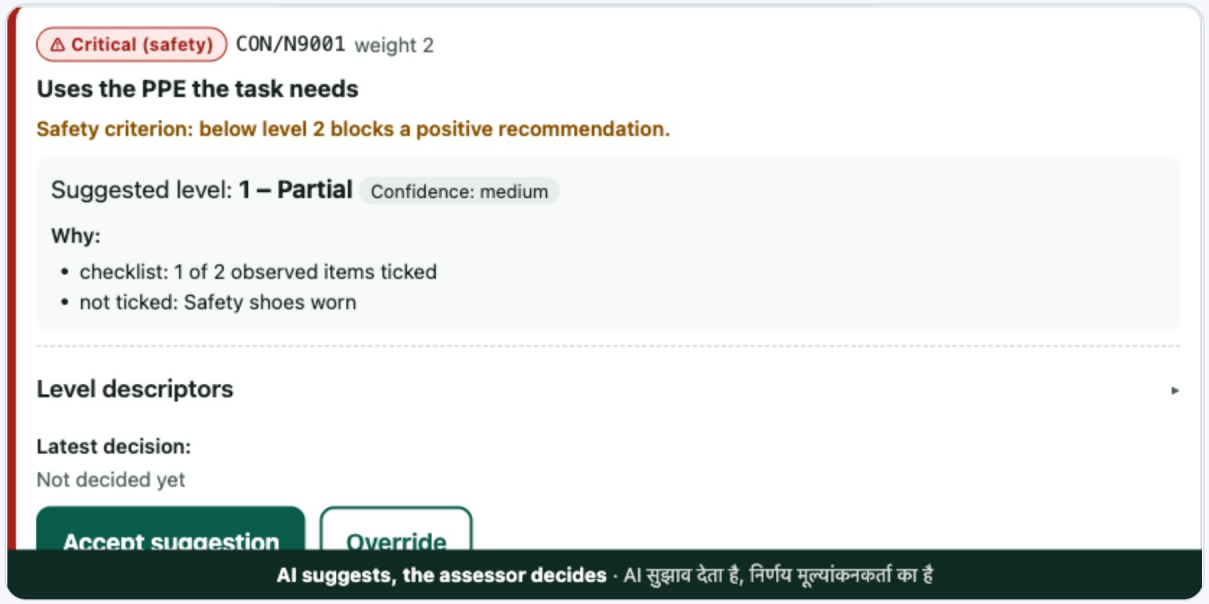
5 qualification files from nqr.gov.in, parsed into 27 skill standards and 344 criteria



Prototype screen, sample data

STEPS 5 AND 6 · SCORE AND DECIDE

Explainable scores, assessor in charge



Prototype screen, sample data

- Four anchored levels (0–3) per criterion, each linked to an official skill-standard code
- Every suggested level shows its reason in plain words
- Official 70% pass mark; a failed safety criterion blocks a positive recommendation
- Overrides need a reason; every decision goes into a hash-chained audit log

STEPS 3 AND 4 · PRACTISE AND CHECK

Evidence you can trust

Consent and privacy

Written consent comes before any photo. Faces are blurred before a photo is stored.

Safety gear check

A YOLOv8n model reports whether helmets, gloves and tools are visible. It never judges workmanship.

Accuracy: RESULT PENDING

Reused-photo check

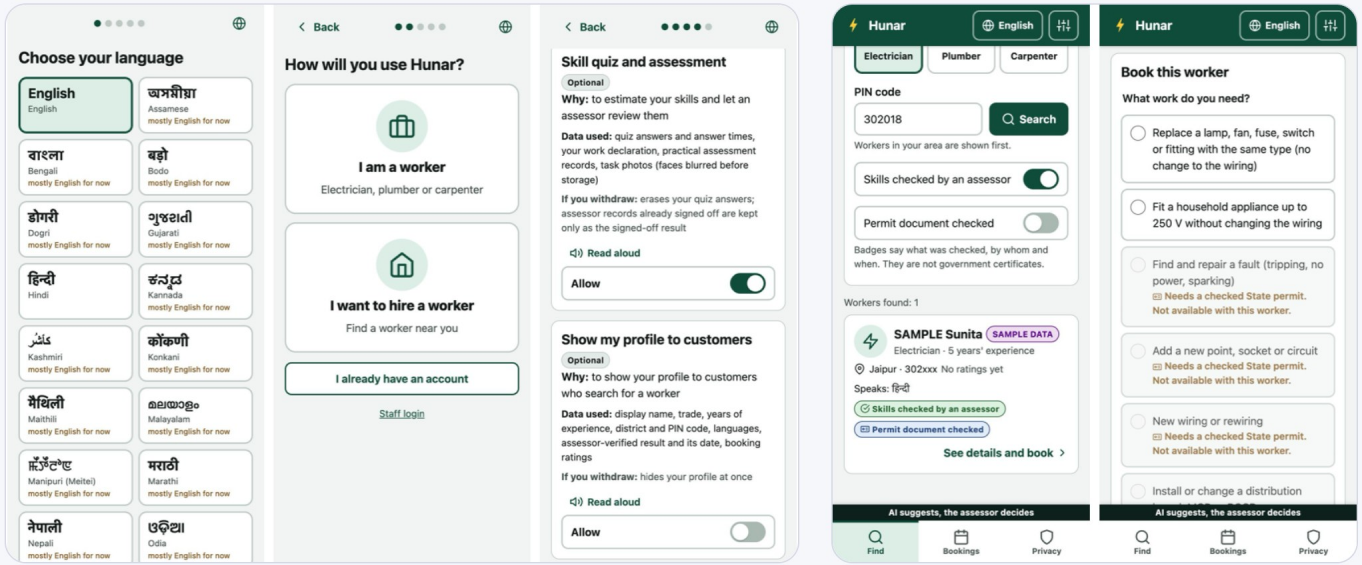
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recall on re-compressed or resized copies, with 1.6% false flags. Crops and rotations are not caught yet.

Reuse test: 162 real PPE photos with simulated re-uploads, 245 fresh photos

BEYOND THE ASSESSMENT

For workers and customers



Prototype screens, sample data

- 23 languages to choose from; English and Hindi complete
- Phone + PIN login, with consent asked for each purpose (DPDP Act 2023)
- Workers: self-check quiz, request an assessment, manage bookings
- Customers: find workers by trade and PIN code; “skills checked” badge only after an assessor signs off

TECHNICAL APPROACH

Technology stack

In the phone

- React + Vite progressive web app
- Offline queue in IndexedDB (Dexie)
- Voice input and read-aloud
- English and Hindi interface

On the server

- Python 3.11 and FastAPI
- SQLModel on SQLite, PostgreSQL-ready
- Hash-chained audit log
- Phone + PIN accounts with staff roles

AI models

- MiniLM and LaBSE multilingual embeddings
- faster-whisper speech to text
- YOLOv8n safety-gear detector
- OpenCV face blur, perceptual hashing

Runs on ordinary CPUs. A GPU is needed only to train the photo model.

FEASIBILITY

Fast enough on an ordinary laptop

0.39 s

to process one photo on one CPU core: decode, face blur, hashing and detection

2.7 ms

to check a new photo for reuse against 1 million stored photos

0.67 s

of CPU time per second of speech, the heaviest step

With 12 photos and 2 minutes of speech per assessment, speech is about 95% of the CPU work, so it is the first part to move to a GPU as usage grows.

Measured with make bench on an Apple M3 laptop; random hashes, synthetic speech

FEASIBILITY

Risks and how we handle them

Risk	How Hunar handles it
The AI suggests something wrong	It only suggests; the assessor decides, and every override is logged with a reason
Photos of real people	Written consent first, faces blurred before storage, consent per purpose (DPDP Act 2023)
Poor internet at assessment sites	Offline-first: actions are queued on the phone and sync in order later
Low literacy, many languages	Voice input, read-aloud and a choice of 23 languages
Fake or reused photo evidence	Every new photo is checked against all stored photos from other workers
Too little real training data	Public data re-split to stop leakage; consented photos of real tasks planned

IMPACT

Who benefits

Workers

On-the-job skill recognised against national standards, opening better work

Assessors

Standard rubrics and explained suggestions, with the final say always theirs

Agencies and skill councils

Comparable results across locations and a tamper-evident record of every decision

Customers

Find nearby workers whose skills an assessor has checked

STATUS AND ROADMAP

Where we are

Built

- Working prototype for Assistant Electrician (CSDCI, NSQF level 3)
- Official data from nqr.gov.in
- Offline app, audit log, worker and customer flows
- Automated test suite

In progress

- Photo detector run 2: RESULT PENDING
- Agreement study with real assessors: RESULT PENDING
- Rubric review by a certified assessor
- Real declarations (mapper tested on synthetic ones so far)

Next

- More trades: plumber, carpenter, mason
- Pilot with an assessment agency
- All 22 scheduled languages
- Speech recognition tuned for Indian accents

RESEARCH

References

1. NCVET, MSDE, “National Skills Qualification Framework-2023,” F. No. 22001/01/2023/NCVET, 2023.
2. NCVET, “Guidelines for Recognition of Prior Learning (RPL),” New Delhi, Aug. 11, 2023.
3. MSDE, “Guidelines for Pradhan Mantri Kaushal Vikas Yojana 4.0,” n.d.
4. National Qualification Register, Government of India, nqr.gov.in (accessed Oct. 28, 2026).
5. NSO, MoSPI, “Annual Report, Periodic Labour Force Survey (PLFS), January–December 2025,” Mar. 2026.
6. Government of India, “The Digital Personal Data Protection Act, 2023,” enacted Aug. 11, 2023.
7. Central Electricity Authority, “(Measures relating to Safety and Electric Supply) Regulations, 2023,” Jun. 8, 2023.
8. N. Reimers and I. Gurevych, “Making monolingual sentence embeddings multilingual using knowledge distillation,” EMNLP, 2020.
9. F. Feng, Y. Yang, D. Cer, N. Arivazhagan and W. Wang, “Language-agnostic BERT sentence embedding,” ACL, 2022.
10. G. Jocher, A. Chaurasia and J. Qiu, “Ultralytics YOLOv8,” version 8.0.0, 2023.
11. A. Radford et al., “Robust speech recognition via large-scale weak supervision,” ICML, 2023.
12. J. L. Fleiss, “Measuring nominal scale agreement among many raters,” Psychological Bulletin, vol. 76, no. 5, 1971.
13. K. Krippendorff, Content Analysis: An Introduction to Its Methodology, 2nd ed. Sage, 2004.